

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) combining a machine learning foundation (Scikit-Learn, Keras, TensorFlow) with hands-on experience building automated, production-style data and AI workflows at Deutsche Bundesbank. Uses Claude Code daily as an AI coding assistant to accelerate development while keeping full ownership of design decisions. Builds auditable, end-to-end pipelines from raw data through to documented, tested output.

Core Competencies

- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means, predictive modeling and clustering
- **AI-Assisted Development:** Daily use of Claude Code as an AI coding assistant to accelerate implementation of data pipelines and automation logic
- **Pipeline Automation:** Configuration-driven, parameterized Python/SQL solutions replacing manual, case-by-case processes at Deutsche Bundesbank
- **Data Engineering:** Automated data-quality testing (Pandas), documentation, and QA-style validation across the pipeline lifecycle
- **Programming:** Python, SQL, R, Java, C++; Git version control
- **Languages:** German (C1), English (C1, TOEFL iBT 105/120), Spanish (native)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations), with automated Jira ticketing and emails to reporting banks
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Automated Validation Framework for Regulatory Reporting Data – converted a non-scalable, case-by-case validation script into a parameterized, reusable Python solution

 - Designed parameterized MySQL queries applicable to any database, attribute, and reporting period, replacing hard-coded per-case script logic
 - Built a YAML-driven configuration interface enabling batch validation across multiple databases, attributes, and reporting periods in a single call

Projects

- **Personal Project (GitHub, built with Claude Code)**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers; 5 quality visualizations and a structured exception report
- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Built predictive models (Random Forest, Neural Networks) and K-Means clustering to identify patient risk profiles from 25+ years of Mexican national health data
- **Tecnológico de Monterrey**
Mexico City Air Quality Analysis – SVM Classification 04/2023–06/2023
 - Applied SVM models (linear & RBF) to classify critical ozone levels from city-wide NO_x/O₃ data with high accuracy

Education

- **Tecnológico de Monterrey** Mexico
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, numerical optimization.
- **Universität Göttingen** Germany
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.